How-To Library

Informatica

Prerequisites to create a Microsoft Azure Synapse SQL connection

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Abstract

You can use Microsoft Azure Synapse SQL Connector to connect to Microsoft Azure Synapse SQL from Cloud Data Integration. This article explains the prerequisite tasks that you must complete before you create a Microsoft Azure Synapse SQL connection.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector

Table of Contents

Overview. 2
Configuring Azure Active Directory authentication. 2
Obtaining the JDBC URL. 3
Obtaining credentials for shared key authentication. 4
Obtaining credentials for service principal authentication. 4
Creating a Microsoft Azure Synapse SQL connection. 10

Overview

You can use Azure Active Directory authentication or Microsoft SQL Server authentication to connect to Microsoft Azure Synapse SQL.

Before you create a Microsoft Azure Synapse SQL connection, complete the following prerequisite tasks:

1. Configure Azure Active Directory (AAD) authentication to connect to Microsoft Azure Synapse SQL.
2. Obtain the JDBC URL or connection string from Microsoft Azure Synapse SQL.
3. Obtain the credentials for shared key authentication and service principal authentication to connect to Azure storage to stage the files:
   - Shared key authentication - Obtain the account name and account key to connect to Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2.
   - Service principal authentication - Applicable to Microsoft Azure Data Lake Storage Gen2. Register an application in the Azure Active Directory, generate a client secret, and then assign the Storage Blob Contributor role to the application.

For more information about Microsoft Azure Synapse SQL Connector, see the *Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector documentation*.

Configuring Azure Active Directory authentication

To configure AAD authentication, perform the following tasks:

Import the server certificate

If a trust store file is not configured for your organization and you want to use AAD authentication with Active Directory Federation Services in Azure, import the server certificate.

Import the server certificate to the following location:

```
<Secure Agent installation directory>\jdk\jre\lib\security\cacerts
```

Use the following command to import the certificate:

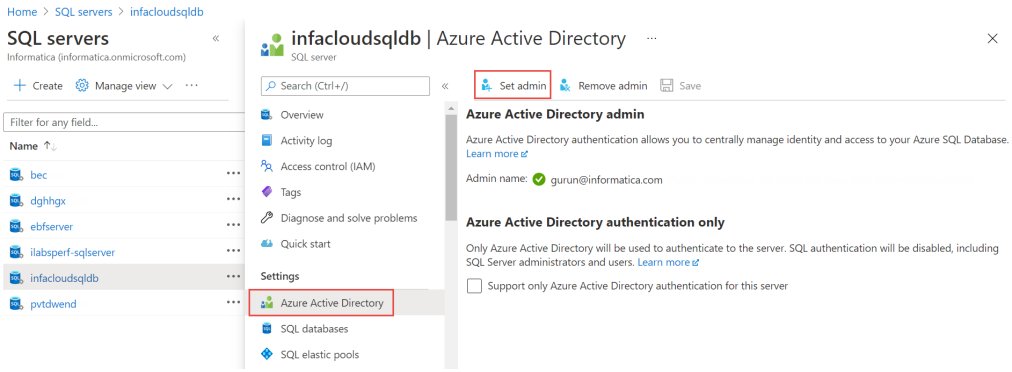
```
keytool -import -trustcacerts -alias <alias name of the certificate> -file <certificate file path> -keystore <Secure Agent installation directory>\jdk\jre\lib\security\cacerts -storepass <password for the truststore>
```

Set the Azure Active Directory admin

To add new users or delete existing users from your Azure Active Directory, you must have the administrator role.

Perform the following steps to set Azure Active Directory administrator:

1. Log in to the Azure portal.
2. On the All Resources page, select the Microsoft SQL Server that hosts Microsoft Azure Synapse SQL.
3. Select the **Azure Active Directory** option under Settings.
   The image shows the Azure Active Directory settings:



4. Click **Set admin**.
   The Add admin page appears.
5. Enter the email ID that you want to use as admin, and click **Select**.
6. Click **Save**.

Create a new user

Perform the following steps to create a user:

1. Connect to Microsoft Azure Synapse SQL using the Azure Active Directory admin created in the previous steps.
   You can use Microsoft SQL Server Management Studio to connect to the Microsoft Azure Synapse SQL.
2. Run the following command:
   ```
   create user [user@foobar.com] from external provider;
   ```
   For more information, see the Microsoft Azure documentation.
3. Assign the required privileges to the user.

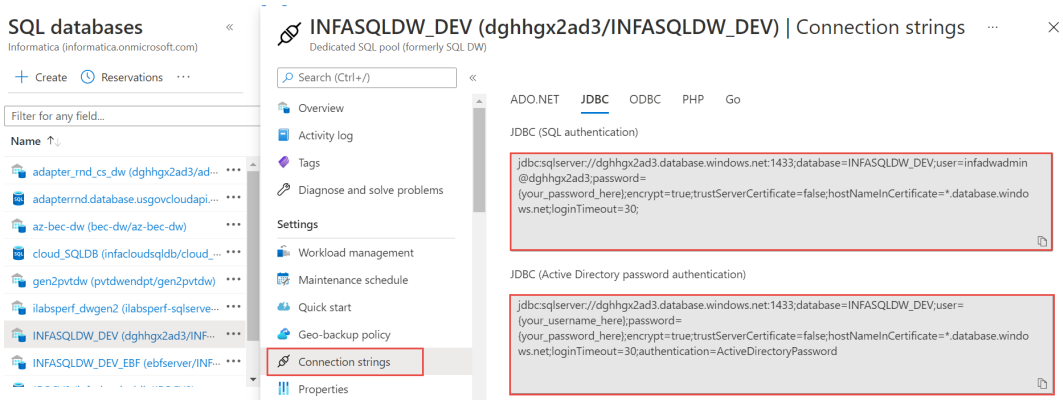
Obtaining the JDBC URL

The JDBC URL contains the credentials to authenticate access to Microsoft Azure Synapse SQL.

To obtain the JDBC URL, perform the following steps:

1. Log in to the Azure portal.
2. Open your Microsoft Azure Synapse SQL account.
3. Under **Settings**, click **Connection strings**.

4. Click the **JDBC** tab and note the connection string based on the type of authentication you want to use to connect to Microsoft Azure Synapse SQL.
   The following image shows the JDBC URL for Microsoft SQL Server authentication and Azure Active Directory authentication:

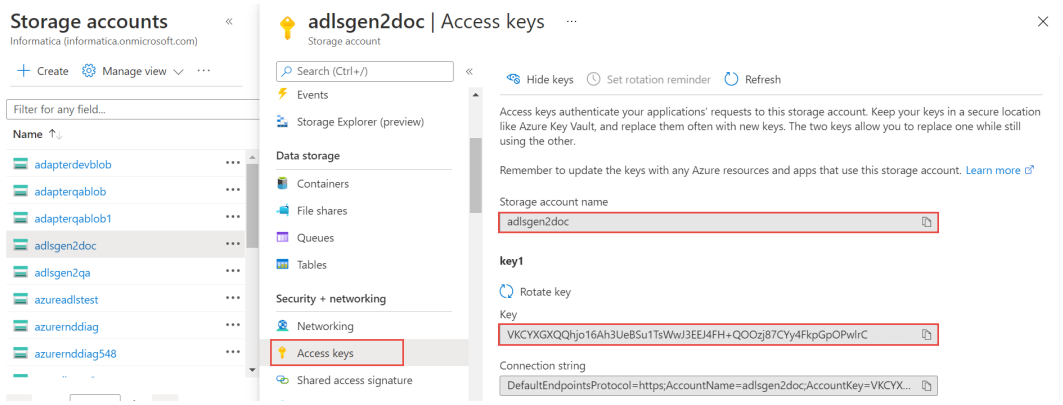


Obtaining credentials for shared key authentication

You can use shared key authentication to connect to Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 to stage files using the account name and account key.

To obtain the account name and account key, perform the following steps:

1. Log in to the Azure portal.
2. Open the storage account.
3. Under **Security + Networking**, click **Access keys**.
4. Click **Show keys** and note the storage account name and account key.



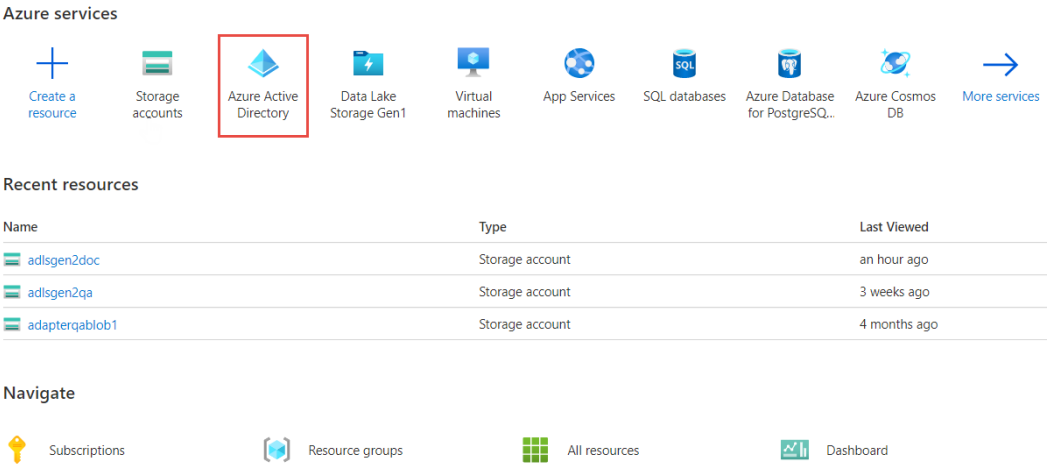
Obtaining credentials for service principal authentication

You can use service principal authentication to connect to Microsoft Azure Data Lake Storage Gen2 to stage files.

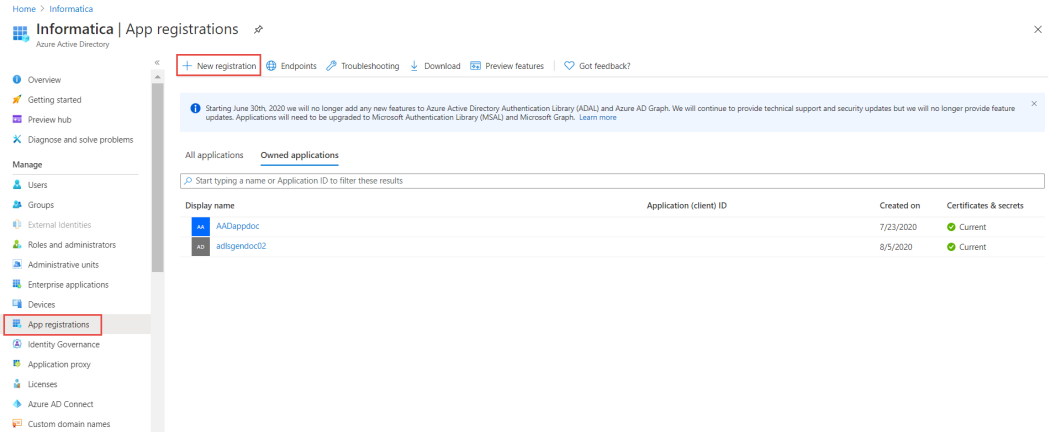
Register an application in the Azure Active Directory, generate a client secret, and then assign the Storage Blob Contributor role to the application.

1. Log in to the Azure portal.

2. Click **Azure Active Directory**.

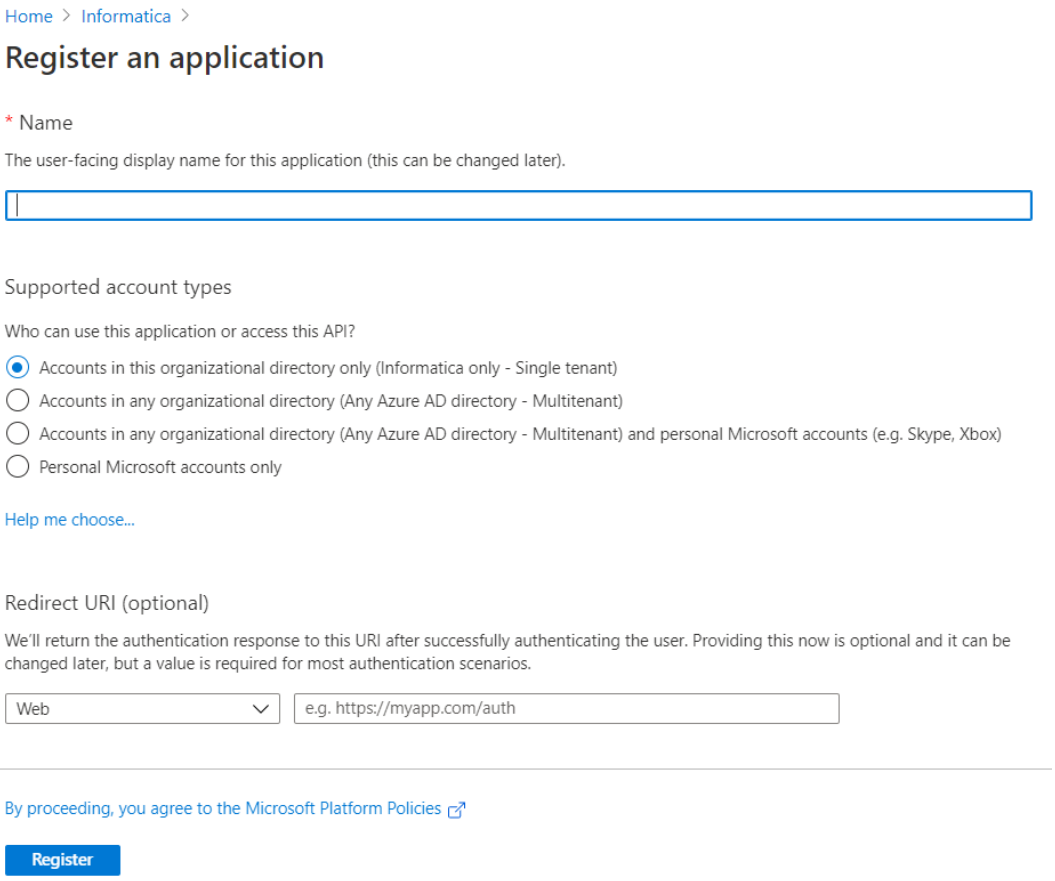


3. In the **Manage** section, click **App registrations**.

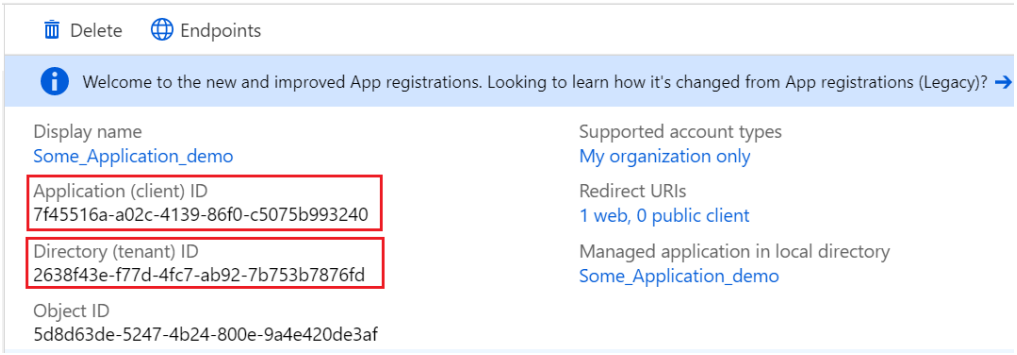


4. Click **New registration** to create a new Azure Active Directory application.

5. On the **Register an application** page, enter the details for the new application.

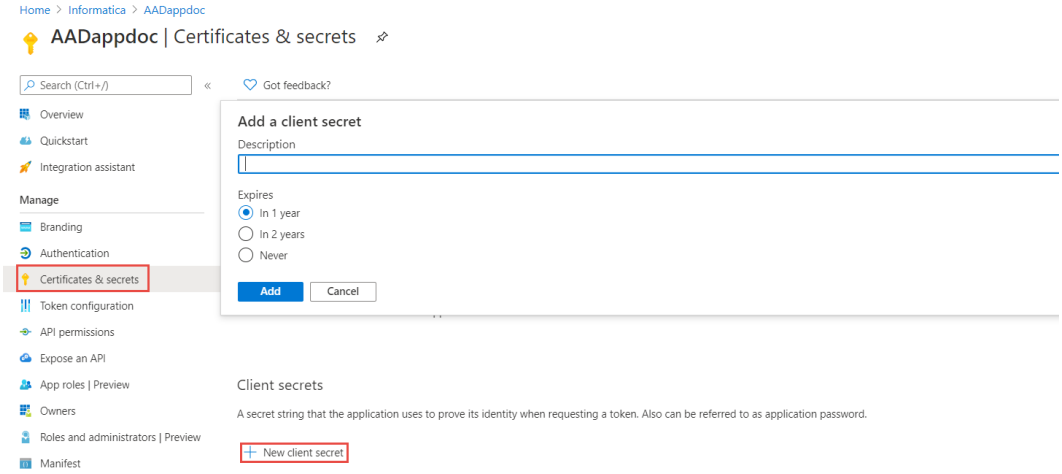


   a. In the **Name** field, enter the application name.
   b. In the **Redirect URI** section, select **Web** as the type of the application and enter the URL of the application.
   c. Click **Register**.
   The details of the newly created Azure Active Directory application page are displayed.

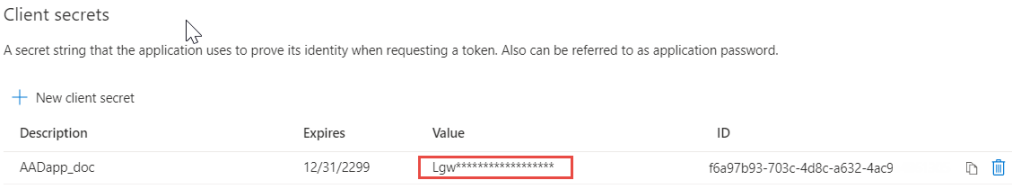


6. In the Manage section, click **Certificates & secrets** section.

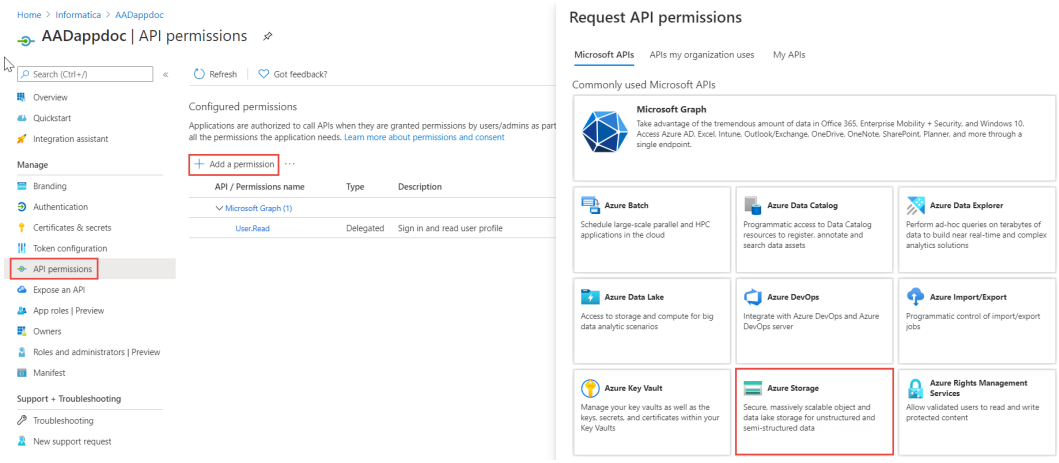
7. Click **New client secret**.



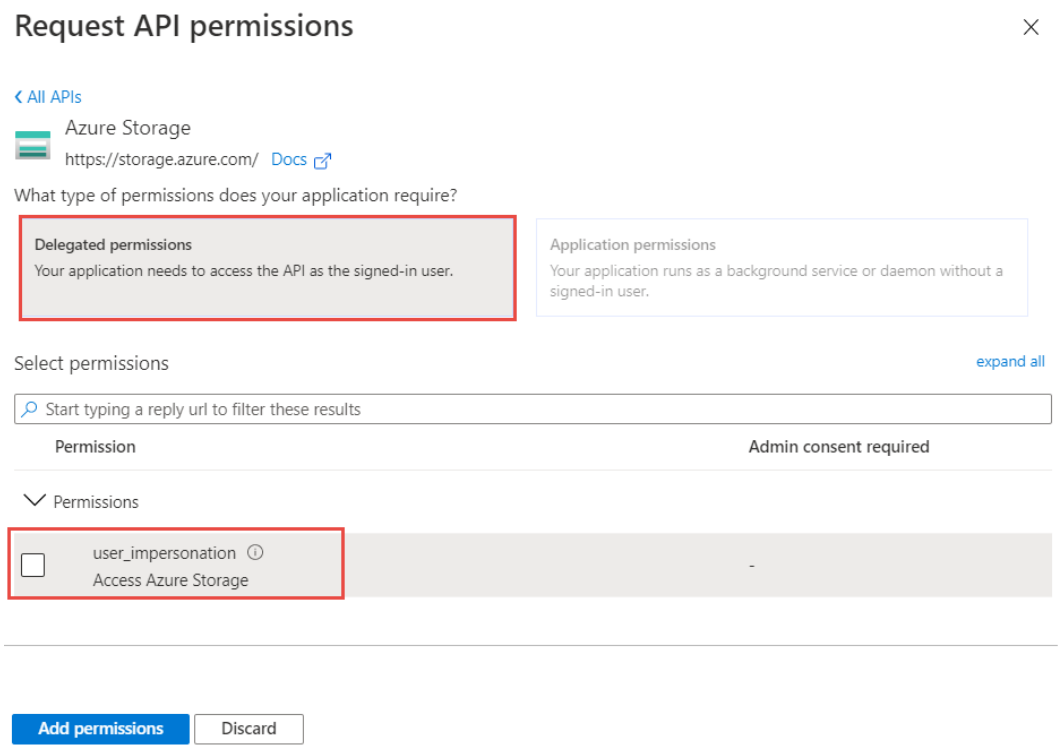
8. In the **Add a client secret** page, perform the following steps:
   a. Enter a name for the client secret in the **Description** field.
   b. In the **Expires** field, you can select the duration of the key as **Never**(Recommended).
   c. Click **Add**.
   d. The value of the key is generated and displayed in the **Value** field.



   Note: You must copy the key value as you cannot retrieve the value after you leave the page. Ensure that the client secret does not contain special characters.
9. In the Manage section, click **Owners**.
10. Click **Add owner**.
11. In the **Search** field, search for the owner name or email address that you used to log in to the Azure portal.
12. Select the owner name or email address, and click **Select**.
13. In the Manage section, click **API permissions**.
    The configured permissions are displayed.



14. Click **Add a permission**.
    The **Request API permissions** page appears.
15. In the Microsoft APIs section, click **Azure Storage**.
16. Select **Delegated permissions** as the type of permissions.
17. Select **Access Azure Storage** from the listed permissions.

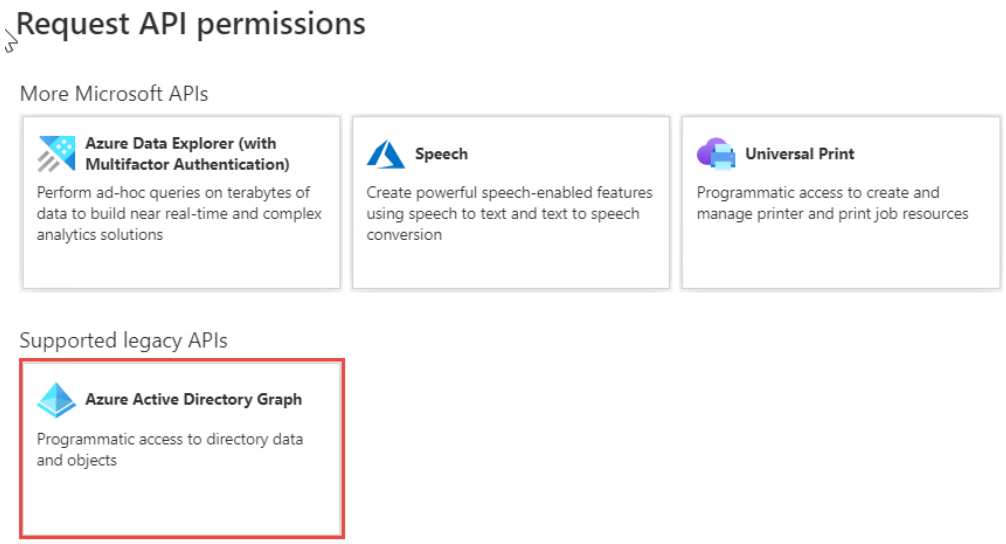


18. Click **Add permissions**.
19. In the **Configured permissions**, select **Azure Active directory** and ensure that the **Sign in and read user profile** option is enabled in the **Delegated permissions** section.

If **Azure Active directory** is not listed under the **Configured permissions**, perform the following steps:

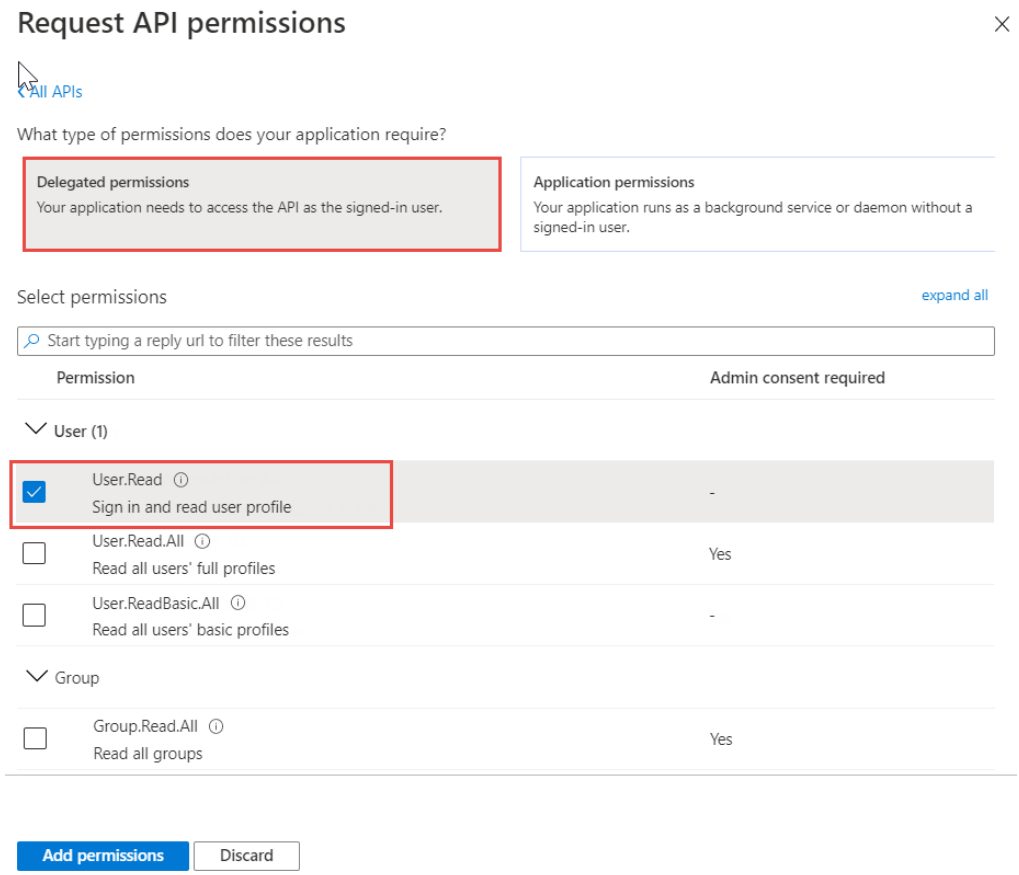
a. Click **Add a permission**.
The **Request API permissions** page appears.

b. In the Microsoft APIs section, click **Azure Active Directory Graph**.



c. Select **Delegated permissions** as the type of permissions.

d. Select **Sign in and read user profile** from the listed permissions.



20. Go to the home page and in the Storage Account section, select the storage account that you created.
21. Click **Access control (IAM)** > **Add**.
22. In the **Add role assignment** page, provide the **Storage Blob Data Contributor** to the application.

Creating a Microsoft Azure Synapse SQL connection

You can use Microsoft SQL Server authentication or Azure Active Directory authentication to connect to Microsoft Azure Synapse SQL. Based on the authentication type, you must provide the JDBC URL in the connection properties.

For information about configuring a Microsoft Azure Synapse SQL connection, see the Informatica Cloud® Data Integration Microsoft Azure Synapse SQL Connector documentation.

After you create the connection, verify the following permissions to perform the read and write operations:

- Verify that either the `db_owner` privilege or the following more granular privileges are granted to the user to connect to Microsoft Azure Synapse SQL and perform read and write operations successfully:
  - `EXEC sp_addrolemember 'db_datareader', '<user>';` // Alternately assign permission to individual table
  - `EXEC sp_addrolemember 'db_datawriter', '<user>';` // Alternately assign permission to individual table
  - `GRANT ALTER ANY EXTERNAL DATA SOURCE TO <user>;`
  - `GRANT ALTER ANY EXTERNAL FILE FORMAT TO <user>;`
  - `GRANT CONTROL TO <user>;`
  - `GRANT CREATE TABLE TO <user>;`
  - Assign required privileges for tasks performed through Pre-SQL and Post-SQL commands.
- Ensure that a default schema is present at the account level or user or group level in Microsoft Azure Synapse SQL.

Author

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